Multilingual Transformers for Low-Resource Fake News Detection

Jordan Samplewood, Riley Demoson
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Abstract

Labelled data for fake news detection is scarce in most languages. We fine-tune multilingual transformers on a low-resource collection covering six languages and show strong transfer from English.

Keywords: multilingual fake news detection, transformers, low-resource languages, misinformation

1. Introduction

Most fake news detectors are built for English. We study how multilingual transformers transfer to languages with little labelled data. This extends the review in A Survey of Fake News Detection Using Machine Learning.

2. Related Work

Multilingual BERT and XLM-R support more than 100 languages. Recurrent approaches such as Recurrent Models for Rumor Detection on Social Media do not transfer well across languages.

3. Methodology

We fine-tune mBERT and XLM-R on English data and evaluate zero-shot and few-shot transfer on five other languages.

4. Results

XLM-R reached 90.4 percent accuracy in the few-shot setting. Zero-shot accuracy was about six points lower.

5. Limitations

Labelled low-resource data is scarce, so evaluation sets are small. The transformers have a high computational cost.

6. Conclusion

Multilingual transformers make fake news detection feasible beyond English.

7. Future Work

Future work should distil these models into efficient versions and cover more languages.

References

[1] A Survey of Fake News Detection Using Machine Learning, Journal of Demonstration Studies, 2021.
[2] Recurrent Models for Rumor Detection on Social Media, Journal of Demonstration Studies, 2020.

Note: this is a SYNTHETIC demonstration paper with fictional authors and invented results, created to demo ResearchGraph. It is not a real publication.